Lecture 1 & 2: GIS and Data Modelling

GIS

- Provide tools for
  - Acquisition
  - Administration
  - Analysis
  - Visualisation

of **spatial Information**

Components:

- Hardware (sensors, computers)
- Software (programs, data)
  - leader in commerical sector: ArcGIS
  - free & open-source: QGIS

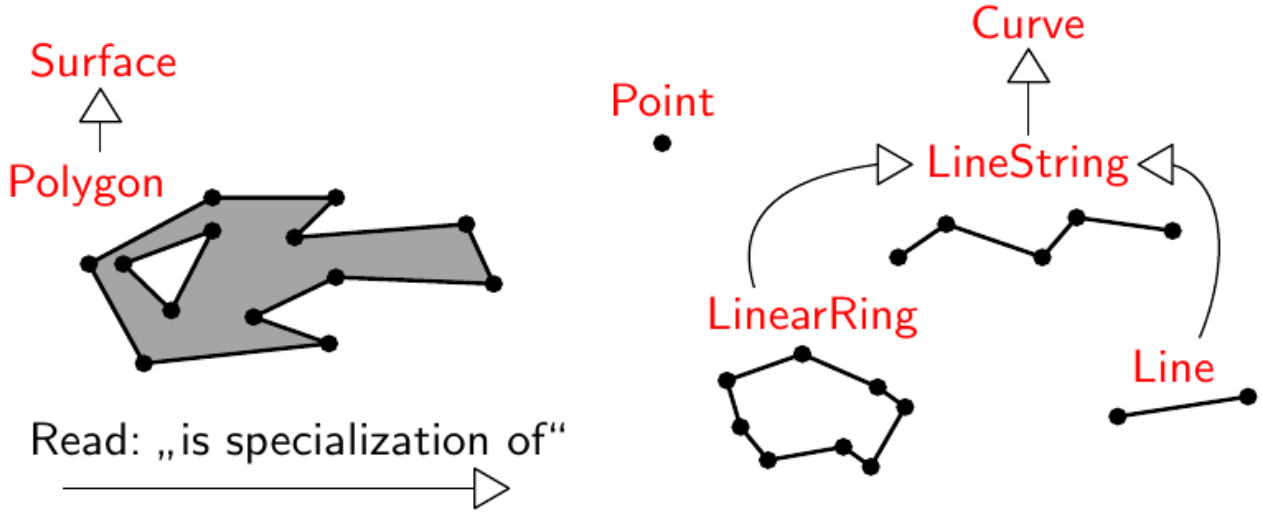
Raster and Vector Data

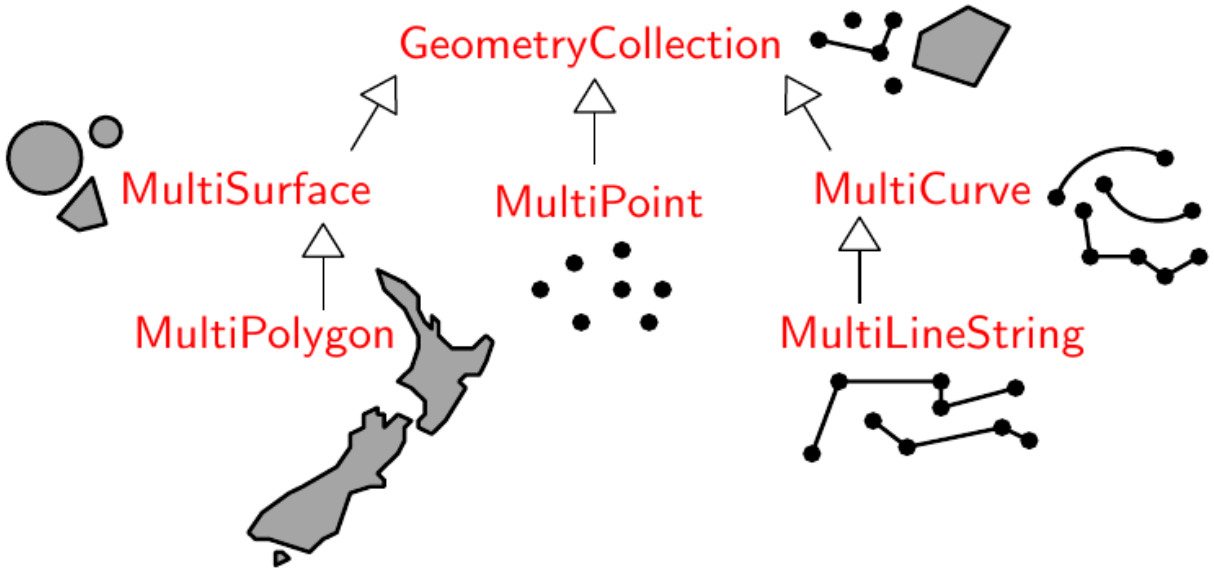
Raster Data

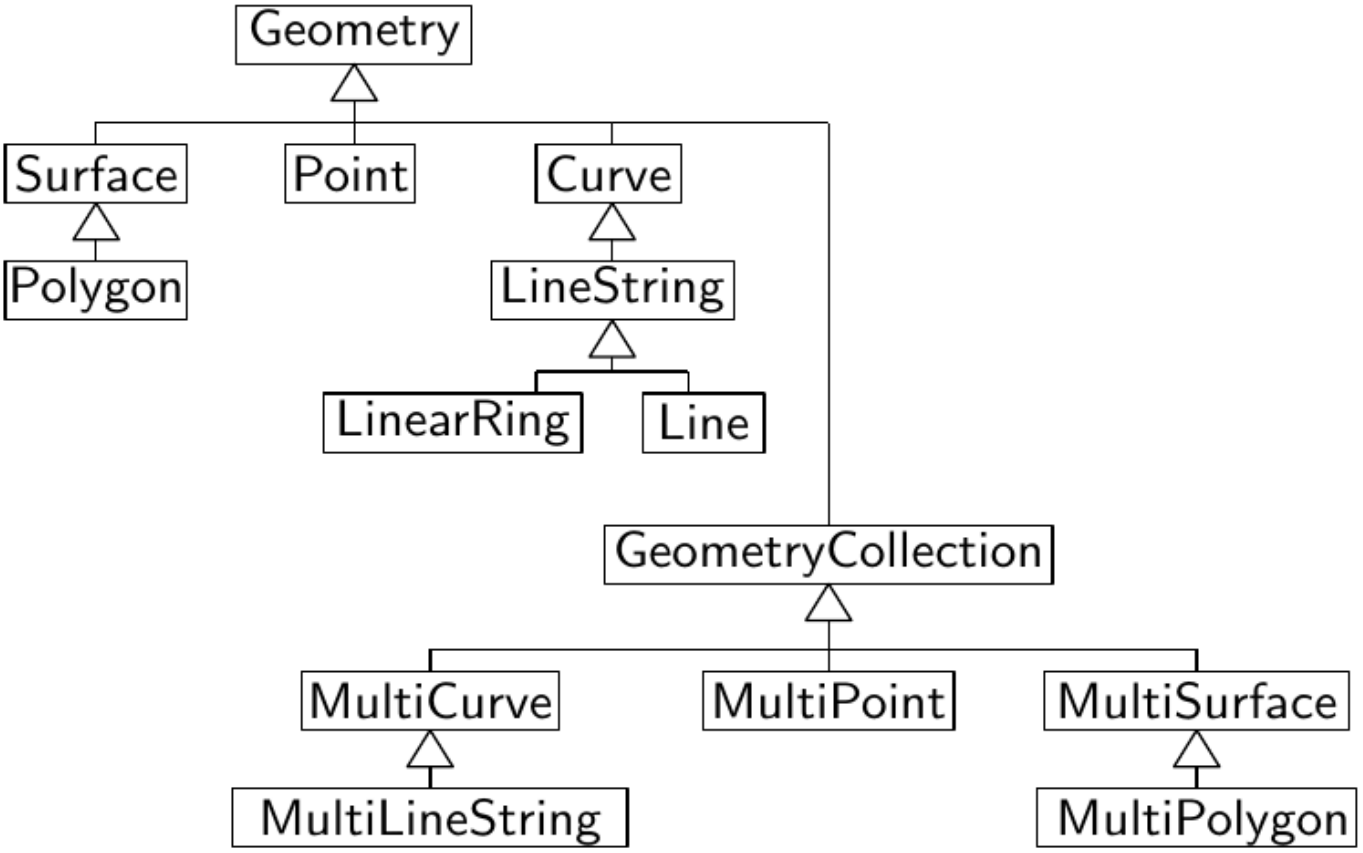
- Used for photos
- 3 channels (r, g, b) for each pixel
  - also other classes (e.g. land-cover class: forest)

Vector Data

- A **geometry** is represented with coordinates
- Geometry types are defined in GIS standard ISO 19125:

- 

  - other meaningful geometry types (not part of standard):
    - CircularArc (Specialisation of Curve)
    - Disk (Specialisation of Surface)
- Multi-Part Geometris:
  - 



WKT Format (Well Known Text)

- Representation of vector geometries as text
- Examples:
  - `POINT(30 10)`
  - `LINESTRING (30 10, 10 30, 40 40)`
  - `POLYGON ((30 10, 40 40, 20 40, 10 20, 30 10))`
  - `POLYGON ((35 10, 45 45, 15 40, 10 20, 35 10),(20 30, 35 35, 30 20, 20 30))`

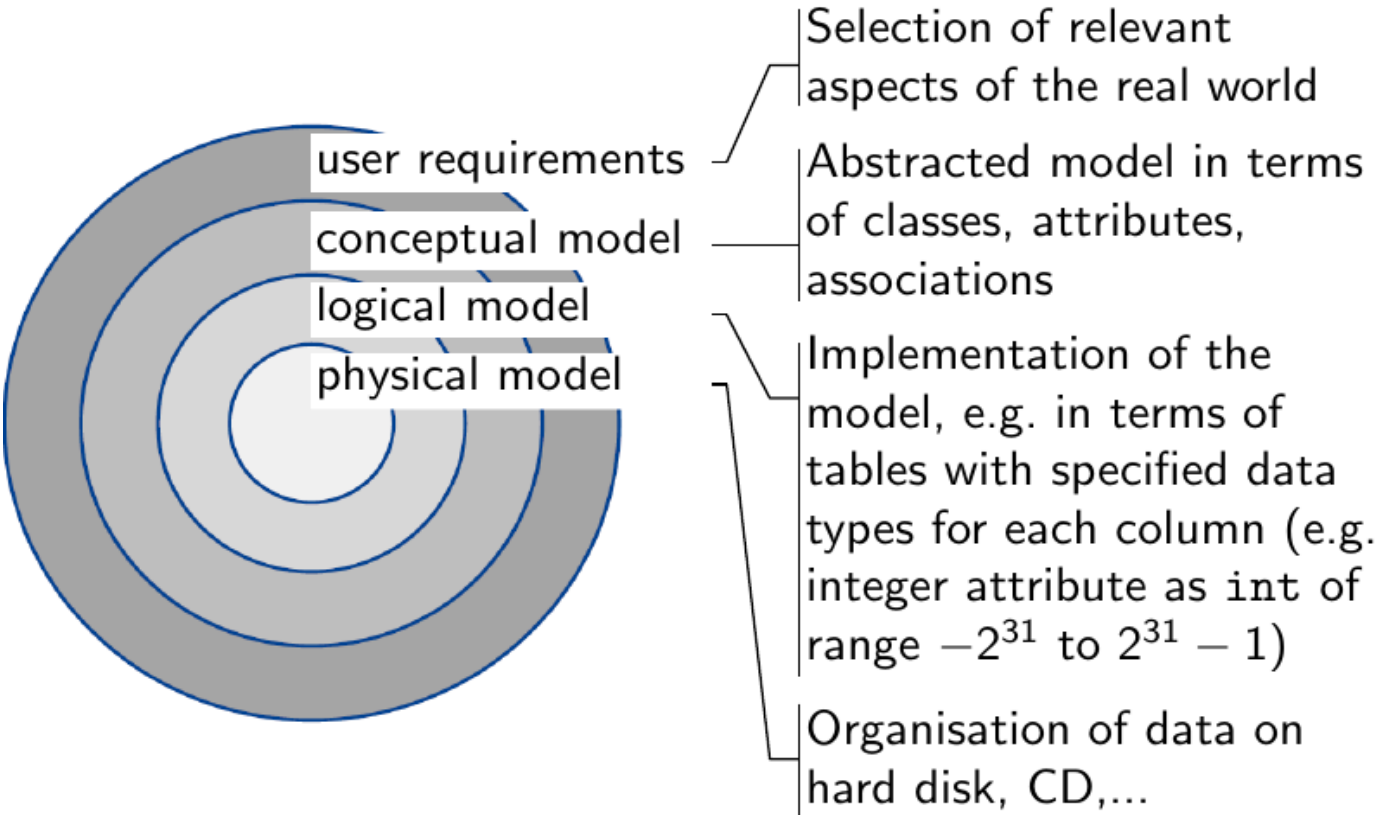
Field-based and object-based view

- Non-spatial attributes, i.e. **semantics**
- Approaches to model real world:
  - **Field**: Each point to value ($f : \mathbb{R}^2 \to \mathbb{R}$)
    - typically in physics: magnet field, gravity field
    - Geographical phenomena: surface height, land-cover type
    - Mathematical terms: function
      - continuous value field: ($f : \mathbb{R}^2 \to \mathbb{R}$)
      - discrete value field/ categorical coverage:($f : \mathbb{R}^2 \to \mathbb{Z}$)
    - With **raster**: pixels mapped to value from a range S
    - With **vector**: partitioned polygons mapped to value from range S

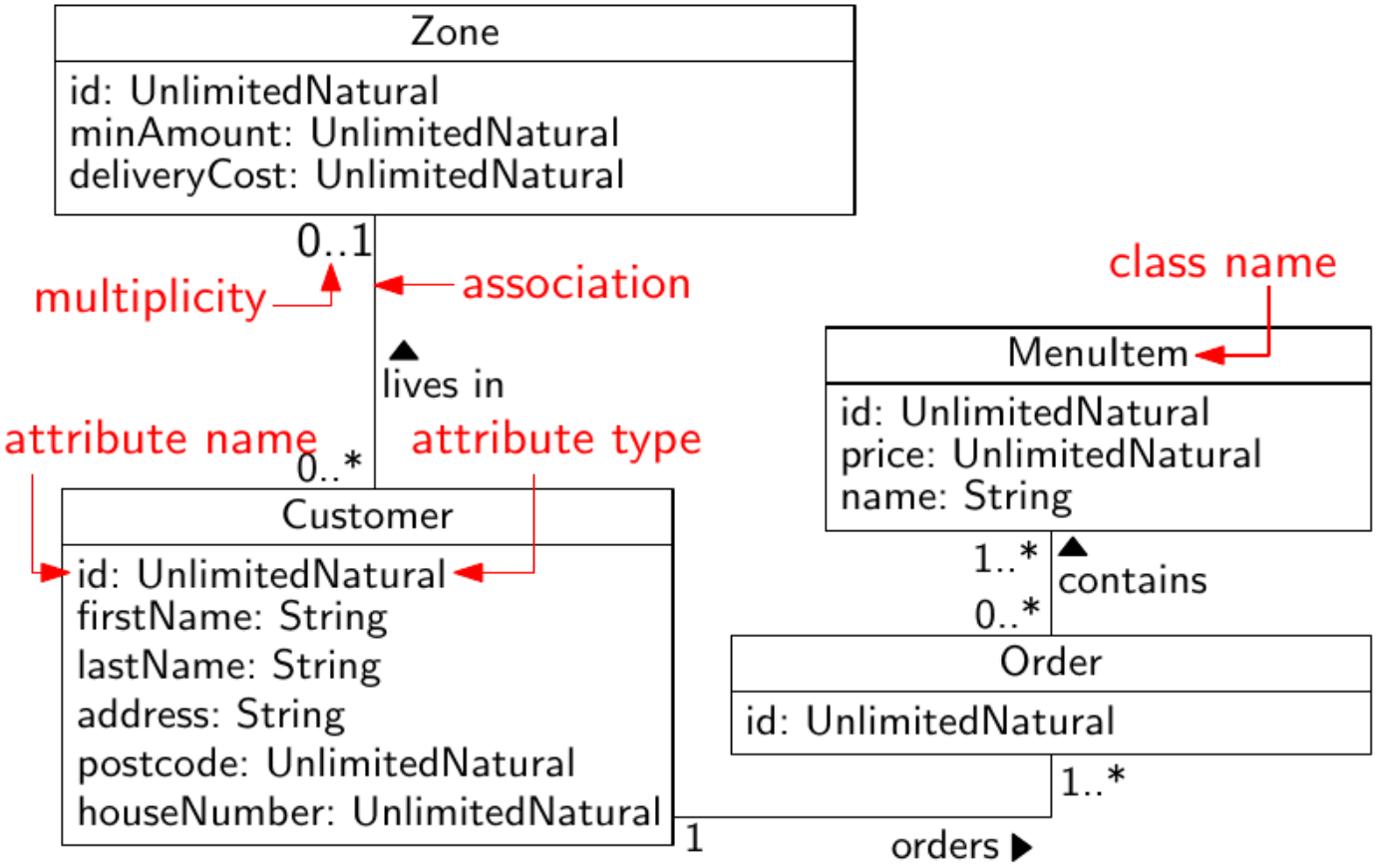
        - Polyguns must not overlap and must not have gaps
        - Boundary between polygons of the same class is not meaningful
        - **All polygons constitue a polygonal subdivision**
    - With **data points**:
        - Set $P = \{p_1, ..., p_n\}$ of points with values $f(p_{1:n})$ given
        - For each other $p \in \mathbb{R}$ a value $f(p)$ can be computed according to **interpolation rule**
  - **Objects**: Discrete real-world entities (buildings, streets) identified and modeled with shapes and attributes
    - *autonomous, unique* object
    - Consists of non-geometrical attributes and a geometry
        - Building A
        - Street X
    - Every object is member of class -> class|object|attribute

Data Modelling

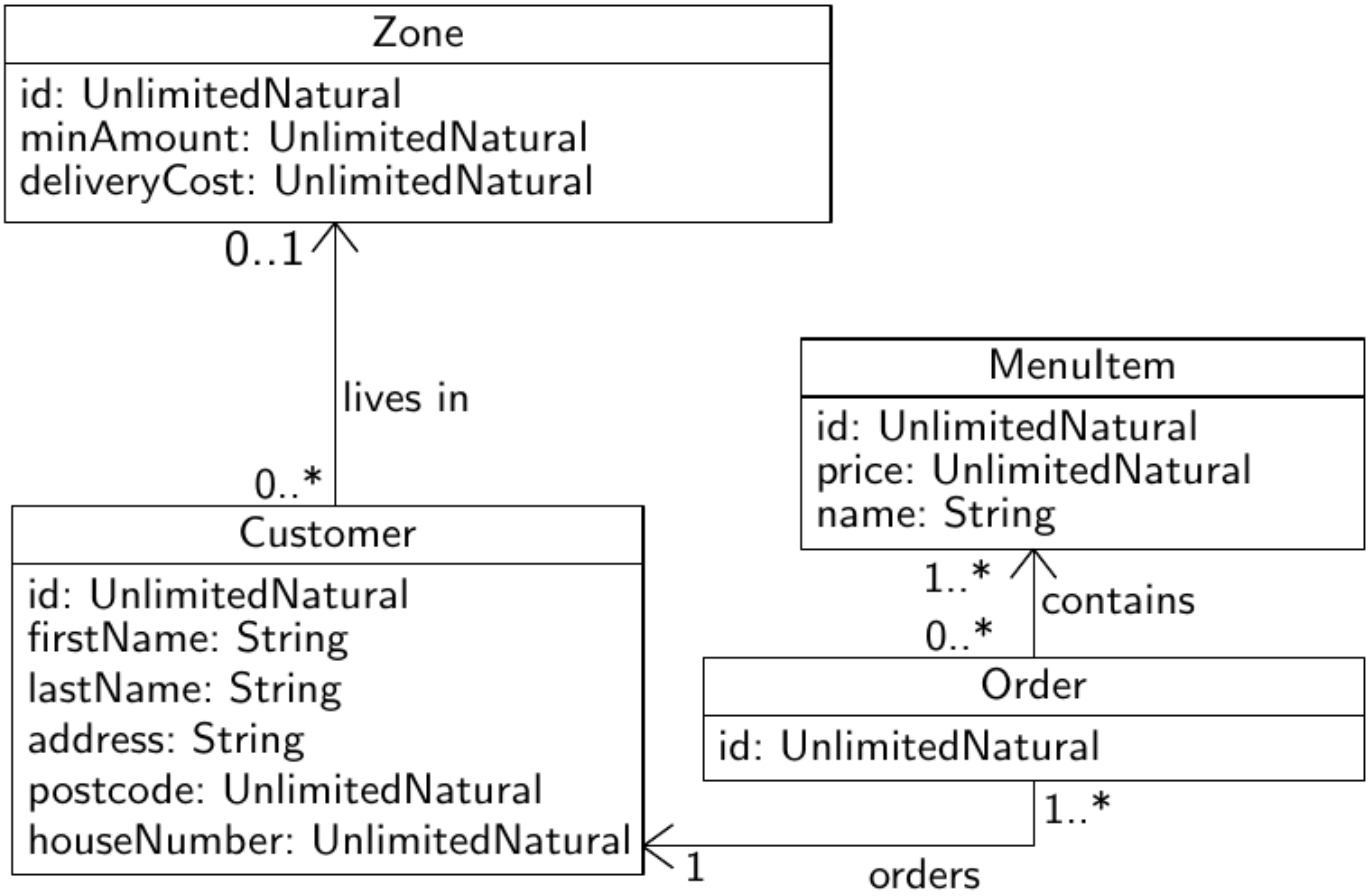
- **Real-world entities** modelled as **objects** (polygons with attributes)
- **Modelling**: abstracting things and selecting certain aspects for a given purpose



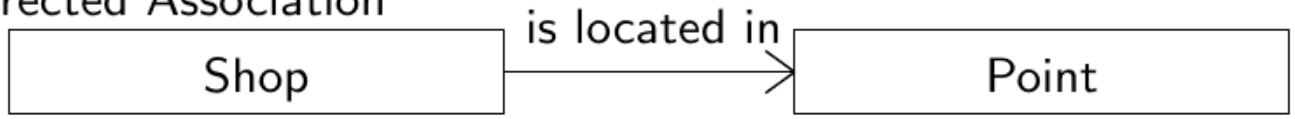
Conceptual Model with UML



Special Associations

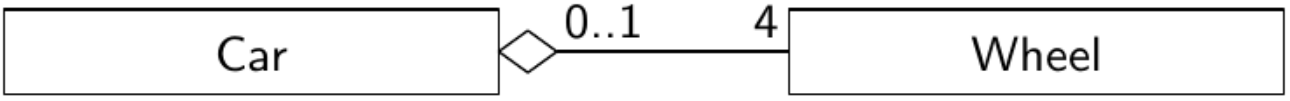


Directed Association



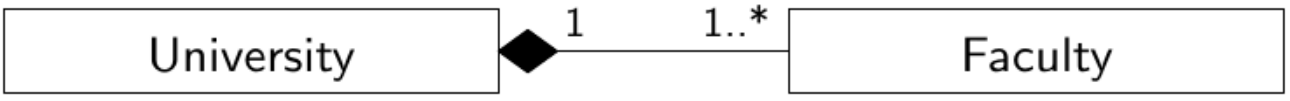
- navigable in only one direction (from Shop to Point)

Aggregation



- a special association meaning "is part of"

Composition



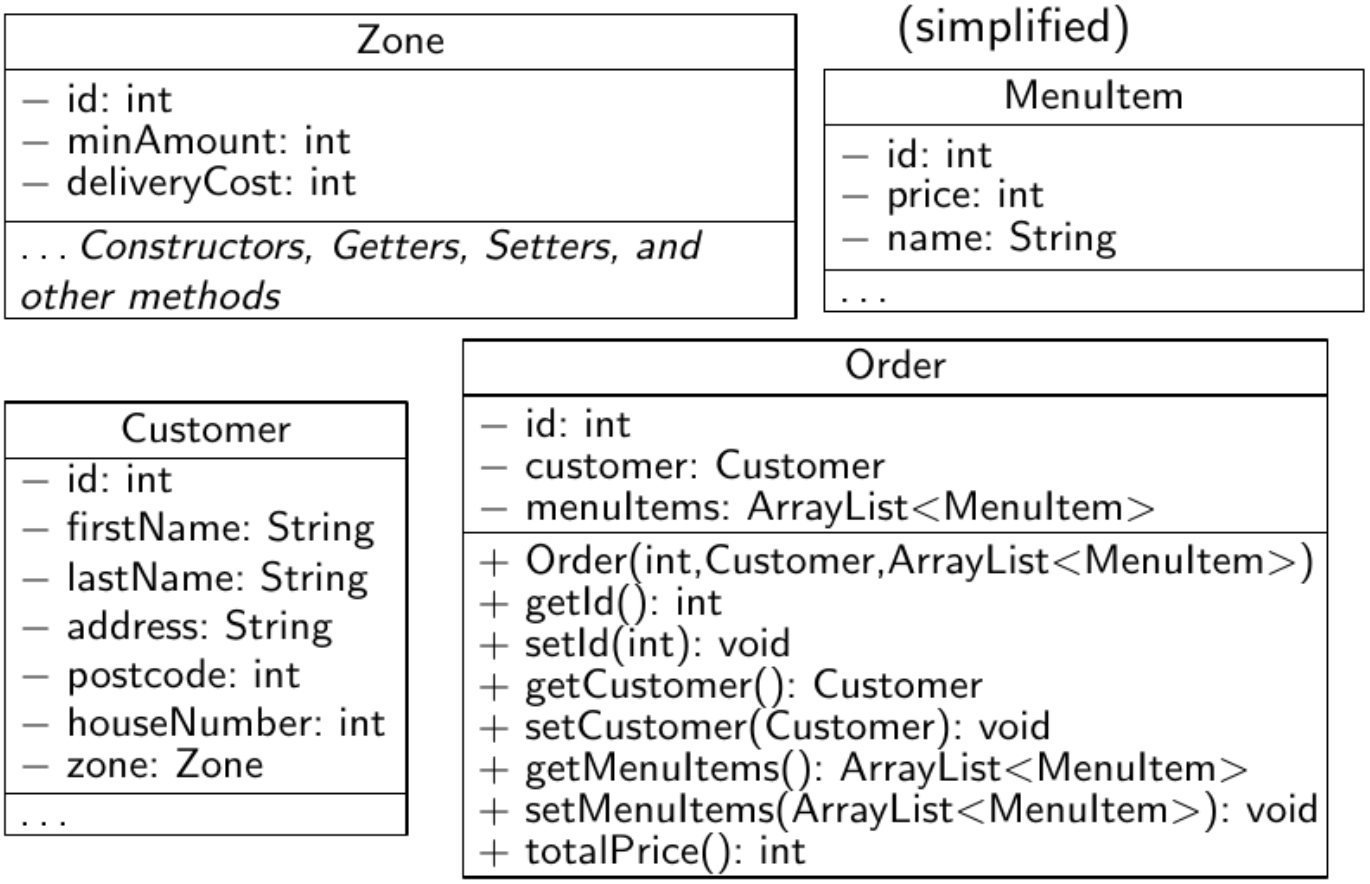
- a special association meaning "is part of" such that the parts (Faculties) cannot exist without the whole (University)

Two possible implementation of the conceptional model are possible:

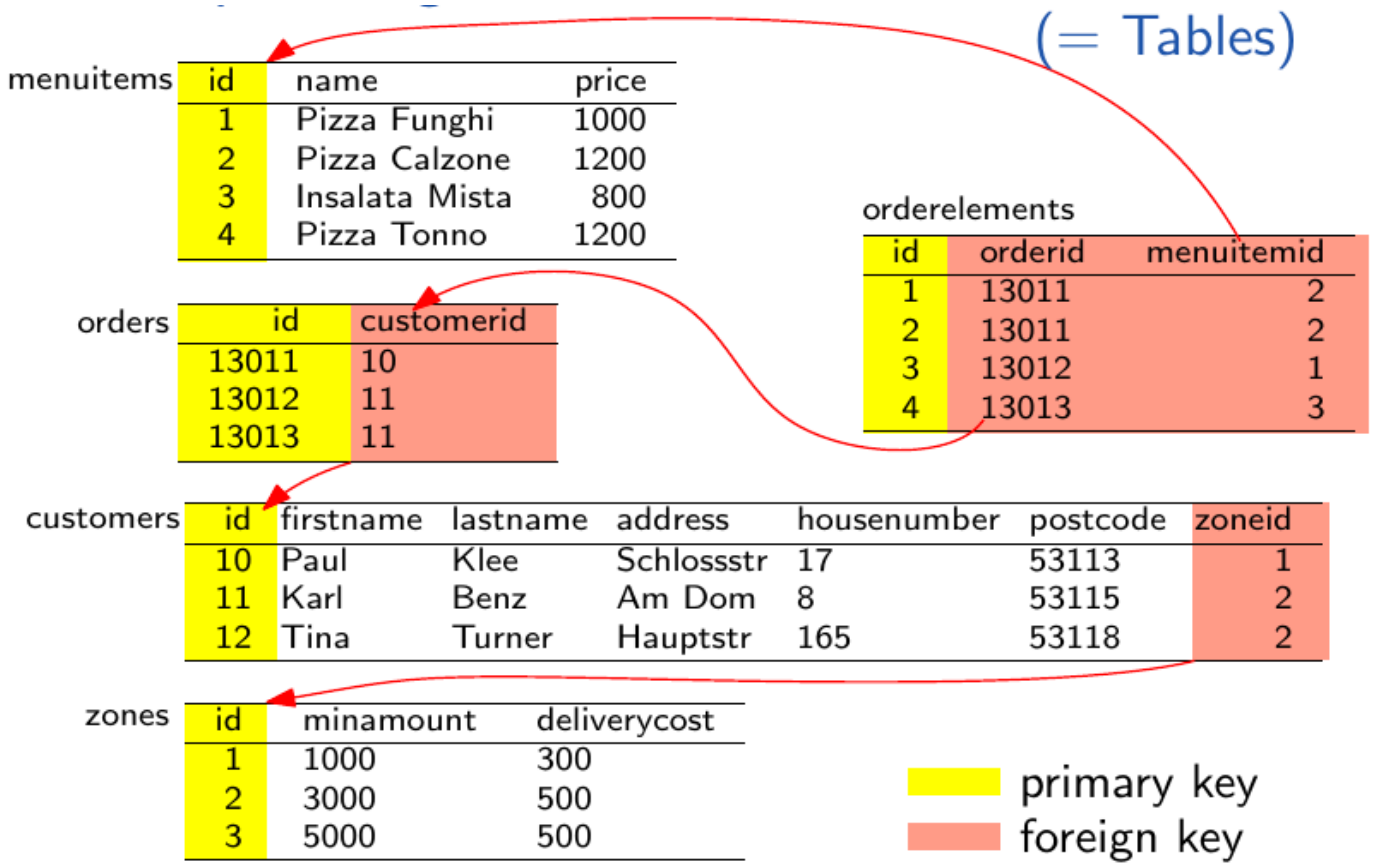
- classes of an object-oriented Java program
- tables of a relational database

Implementation of a Conceptual Model as a Java Program

Attribute types (Concpetual Model)	Java Data types (logical model)
unlimitedNatural, $\mathbb{N}$	int
Real, $\mathbb{R}$	double
String	String



Implementation of a Conceptual Model as a Database



Files vs. Databases

- **Separate Files**:
  - One file (multiple files) for each table
  - problem: **logical integrity** can be violated
    - i.e. data are logically correct and do not contradict:
      - e.g. Key that does not exist
      - concurrent usage
- **Databases**:
  - Interaction via database management system (DBMS)
  - Responsible for
    - Efficient execution of queries
    - Ensuring logical consistency after each transaction

Queries with SQL

WHERE

Which customers live in the street whose name is Schlossallee?

```
SELECT * FROM customers WHERE address = 'Schlossallee';
```

all columns — table — requirement for selected rows

ORDER BY

Return one the most expensive meals.

```
SELECT * FROM menuitems ORDER BY price DESC LIMIT 1;
```

descending order — only first row

Aggregate Functions

Return the highest prize among all meals.

```
SELECT MAX(price) FROM menuitems;
```

aggregate function

AVG()	average
COUNT()	total number
FIRST()	first value
LAST()	last value
MAX()	maximum
MIN()	minimum
SUM()	sum

GROUP BY

For each meal that has ever been ordered, return the id of the first order in which that meal occurs.

```
SELECT menuitemid, MIN(orderid) FROM orderelements
GROUP BY menuitemid;
```

minimum is computed separately for each group

NESTING

Return all the most expensive meals.

```
SELECT * FROM menuitems
WHERE price = (SELECT MAX(price) FROM menuitems);
```

SQL queries can be nested!